be in increasing order).

2.12.5.2. EnergyMeasurementStruct Type

This struct SHALL indicate the amount of energy measured during a given measurement period.

A server which does not have the ability to determine the time in UTC, or has not yet done so, SHALL use the system time fields to specify the measurement period and observation times.

A server which has determined the time in UTC SHALL use the timestamp fields to specify the measurement period. Such a server MAY also include the systime fields to indicate how many seconds had passed since boot for a given timestamp; this allows for client-side resolution of UTC time for previous reports that only included systime.

Elements using this data type SHALL indicate whether it represents cumulative or periodic energy, e.g. in the name or in the element description.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Energy**	energy-mWh	0 to 2^{62}				M
1	**StartTimestamp**	epoch-s					desc
2	**EndTimestamp**	epoch-s	min (StartTimestamp + 1)				desc
3	**StartSystime**	systime-ms					desc
4	**EndSystime**	systime-ms	min (StartSystime + 1)				desc
5	**ApparentEnergy**	energy-mVAh	0 to 2^{62}				P, APPE
6	**ReactiveEnergy**	energy-mVARh	0 to 2^{62}				P, REAE

2.12.5.2.1. Energy Field

This field SHALL be the reported energy.

If the EnergyMeasurementStruct represents cumulative energy, then this SHALL represent the cumulative energy recorded at either the value of the EndTimestamp field or the value of the EndSystime field, or both.

If the EnergyMeasurementStruct represents periodic energy, then this SHALL represent the energy recorded during the period specified by either the StartTimestamp and EndTimestamp fields, the